

Solution Requirements

Date	04 July 2026
Team ID	
Project Name	A Comprehensive Measure of Well-Being
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	No user login is required. The system is publicly accessible for prediction.	Low
2	Authorization levels	All users have the same access to enter data and generate HDI predictions.	Low
3	External interfaces	The system uses the trained HDI.pkl model and Flask framework.	Medium

		No external APIs are required.	
4	Transactions processing	The system accepts user inputs, processes them using the Linear Regression model, and displays the predicted HDI and development category.	High
5	Reporting	The system displays the predicted HDI value and corresponding development category on the result page.	High
6	Business rules, etc.	The prediction is based on Life Expectancy, Expected Years of Schooling, Mean Years of Schooling, and GNI per Capita. The predicted value is classified into Low, Medium, High, or Very High Human Development.	High
7	Compliance to laws or regulations	The application does not collect or store personal information. It follows basic data privacy practices.	Medium
8	Other	The application provides a Predict Again option, allowing users to perform multiple predictions without restarting the system.	Medium

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	The application should generate predictions quickly after user input.	Prediction displayed within 2 seconds
2	Scalability	The system should support multiple prediction requests without affecting performance.	Supports multiple users with minimal delay
3	Security & Data Privacy	User input is processed only for prediction and is not permanently stored.	No sensitive user data is stored
4	Reliability & Availability	The application should provide consistent and accurate predictions whenever it is running.	99% availability during usage
5	Usability & Accessibility	The interface should be simple, responsive, and easy to understand for all users.	User-friendly interface with clear navigation
6	Other	The system should be maintainable and easy to update with new datasets or improved machine learning models.	Easy model replacement and code maintenance